# Efficient CIFAR-100 Modelling

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**Abstract**

I study both discriminative and generative modelling on the CIFAR-100 dataset under strict constraints on model size and training length. For classification, I design a compact convolutional network with approximately 97,000 parameters and train it for 10,000 optimisation steps, achieving a test accuracy of 0.613. For generation, I train a conditional generative adversarial network with approximately 970,000 parameters for 50,000 optimisation steps, producing diverse class-conditional samples with a Fréchet Inception Distance of 28.6. Throughout, architectural and optimisation choices are guided by practical trade-offs between capacity, stability, and computational budget. The results demonstrate that careful design allows both discriminative and generative models to perform reasonably well under tight constraints.

## Part 1: Classification

### Methodology

I formulate CIFAR-100 classification as a supervised learning problem mapping an input image $x \in \mathbb{R}^{3\times32\times32}$ to one of 100 discrete class labels. The network outputs unnormalised class scores (logits) $z \in \mathbb{R}^{100}$, which are converted to class probabilities using the softmax function. Training minimises cross-entropy loss with label smoothing [14], defined over a minibatch as

$$\mathcal{L} = -\sum_k \tilde{y}_k \log p(y = k \mid x),$$

where $\tilde{y}$ denotes the smoothed target distribution with smoothing factor $\varepsilon = 0.1$.

The model is a compact convolutional network built from MobileNetV2-style inverted residual blocks [13] augmented with Squeeze-and-Excitation (SE) [5] and residual connections [3]. The architecture begins with a $3 \times 3$ convolutional stem mapping RGB inputs to 26 channels at full spatial resolution. This is followed by four stages of MBConv blocks. Each block consists of a $1 \times 1$ pointwise expansion, a $3 \times 3$ depthwise convolution [4], channel-wise SE reweighting based on global average pooled features, and a $1 \times 1$ pointwise projection. Residual connections are applied when spatial resolution and channel dimensionality are unchanged.

Downsampling is performed by using stride-2 depthwise convolutions at the start of stages 2–4, reducing the spatial resolution from $32 \times 32$ to $16 \times 16$, $8 \times 8$, and $4 \times 4$, while increasing the number of channels from 26 to 36, 52, and 72 respectively. Batch normalisation and SiLU activations are applied throughout. After the final stage, global average pooling [8] produces a 72-dimensional feature vector, followed by dropout with probability 0.2 and a linear classifier mapping to 100 class logits.

The total parameter count of the model is 97,662. All parameters are trained from scratch. Training data is augmented using random cropping with padding, horizontal flipping, and colour jitter, followed by per-channel normalisation. Test images are normalised only. Optimisation is performed using stochastic gradient descent with momentum [3] and weight decay $5 \times 10^{-4}$. The learning rate is initialised at 0.2 and annealed using a cosine schedule [9] over the full training horizon. Training is carried out for exactly 10,000 optimisation steps with a batch size of 128.

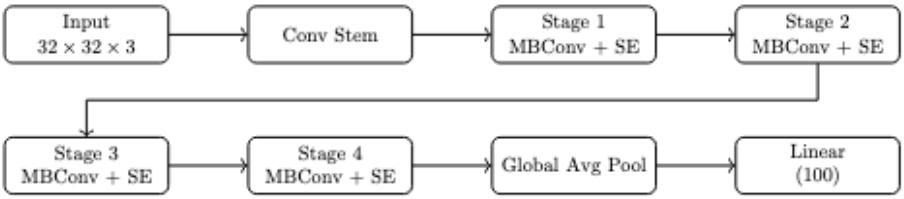



Figure 1: Overview of the discriminative model architecture. The network progresses from a convolutional stem through four MBConv + SE stages, followed by global average pooling and a linear classification head.

## Results

After 10,000 optimisation steps, the model reaches a mean training accuracy of $0.597 \pm 0.044$ and a mean test accuracy of $0.613 \pm 0.043$. The final mean training loss is 2.085. Figure 2 shows the evolution of training and test accuracy over the course of optimisation.

Both training and test accuracy increase smoothly throughout training, indicating stable optimisation with no signs of divergence. The test accuracy slightly exceeds the training accuracy at convergence, resulting in a small negative generalisation gap. This behaviour is consistent with the use of strong regularisation, including data augmentation and label smoothing, which reduce apparent training accuracy while improving robustness on unseen data.

Given the limited model capacity and fixed training duration, the achieved test accuracy is reasonable. CIFAR-100 presents a challenging fine-grained classification task, and achieving test accuracy above 60% with fewer than 100k parameters suggests that the architectural and optimisation choices make effective use of the available capacity.

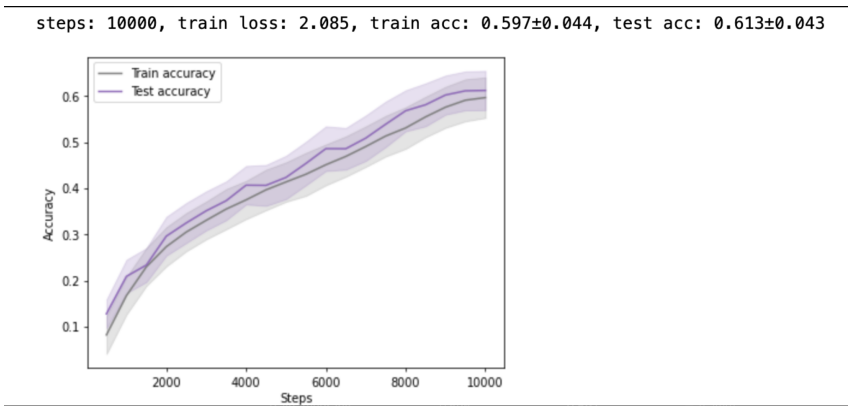



Figure 2: Training and test accuracy curves over 10,000 optimisation steps. Shaded regions indicate standard deviation across batches.

## Limitations

The primary limitation of this model is its restricted representational capacity. While MBConv blocks and SE attention improve parameter efficiency, the shallow depth and narrow channel widths limit the complexity of features that can be learned, particularly for visually similar CIFAR-100 classes.

Generalisation performance appears close to saturation by the end of training, with accuracy curves flattening in later stages. This suggests that additional optimisation steps or modest increases in capacity could yield further improvements, but such changes are not explored here. The cosine learning-rate schedule completes its decay by the end of training, leaving limited opportunity for continued refinement.

Finally, the architecture prioritises simplicity and efficiency over flexibility. Fixed expansion ratios, uniform block structures, and the absence of explicit spatial attention mechanisms constrain architectural diversity. While these choices support stable training under tight capacity limits, they may prevent the model from capturing finer spatial detail present in the data.

# Part 2: Generative Model

## Methodology

The generative model consists of a generator ($G$) and a discriminator ($D$) trained jointly using adversarial optimisation [1, 10]. The generator defines a mapping

$$x = G(z, y),$$

where $z \in \mathbb{R}^{128}$ is sampled from a standard normal distribution and $y \in \{1, \ldots, 100\}$ denotes the class label. Image synthesis at inference time is performed solely by sampling $z$ and $y$, with no dependence on real images.

**Generator architecture** The generator begins with a fully connected layer that projects the latent vector into a low-resolution spatial feature map. This representation is progressively upsampled to the target image resolution through a sequence of residual blocks [2]. Each block applies conditional batch normalisation, non-linear activation, and convolution, with skip connections that preserve information across scales and support stable gradient flow.

Class conditioning is implemented through conditional batch normalisation, where each class label controls learned scale and bias parameters applied to intermediate feature maps [15]. This mechanism allows class-specific modulation while sharing the core convolutional structure across all categories. A self-attention module is included at an intermediate spatial resolution to capture longer-range dependencies that cannot be modelled effectively by local convolutions alone [16]. The final layer maps features to RGB space and applies a hyperbolic tangent activation to produce normalised images.

**Discriminator and conditioning** The discriminator follows a residual downsampling architecture. Spectral normalisation is applied to all learnable layers to regularise the discriminator and stabilise adversarial training [12]. Class conditioning is incorporated through a projection mechanism, in which an embedding of the class label interacts with the final feature representation before producing a scalar score [11]. The discriminator is used only to define the adversarial objective and is not involved in sampling.

**Optimisation objective and training procedure** Training uses the hinge adversarial loss [7]. The discriminator objective is

$$\mathcal{L}_D = \mathbb{E}_x[\max(0,\, 1 - D(x, y))] + \mathbb{E}_z[\max(0,\, 1 + D(G(z, y), y))]\,,$$

while the generator is optimised using

$$\mathcal{L}_G = -\mathbb{E}_z[D(G(z, y), y)]\,.$$

Both networks are optimised using Adam with fixed hyperparameters. Generator and discriminator updates are alternated at each optimisation step. An exponential moving average of the generator parameters is maintained throughout training and used for sampling, which improves consistency and reduces sensitivity to short-term optimisation noise [6].

The combined parameter count of the generator and discriminator is below one million, and training is performed for a fixed total of 50,000 optimisation steps. Standard data augmentation is applied to real images during training to improve robustness, while no external data or pretrained components are used.

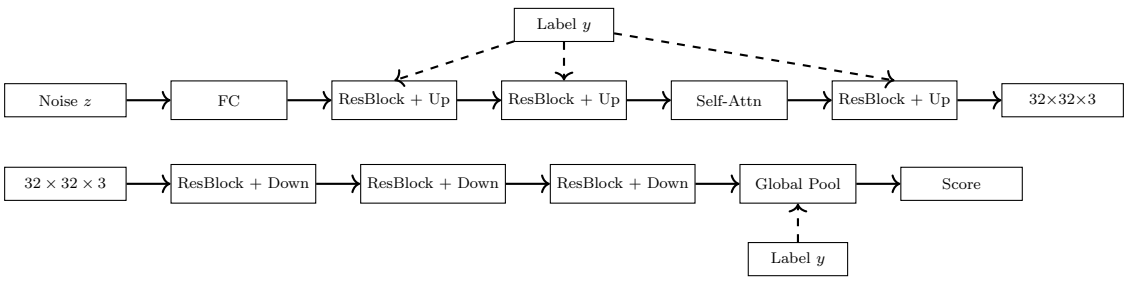



Figure 3: Conditional GAN architecture with residual upsampling in the generator and projection-based discrimination.

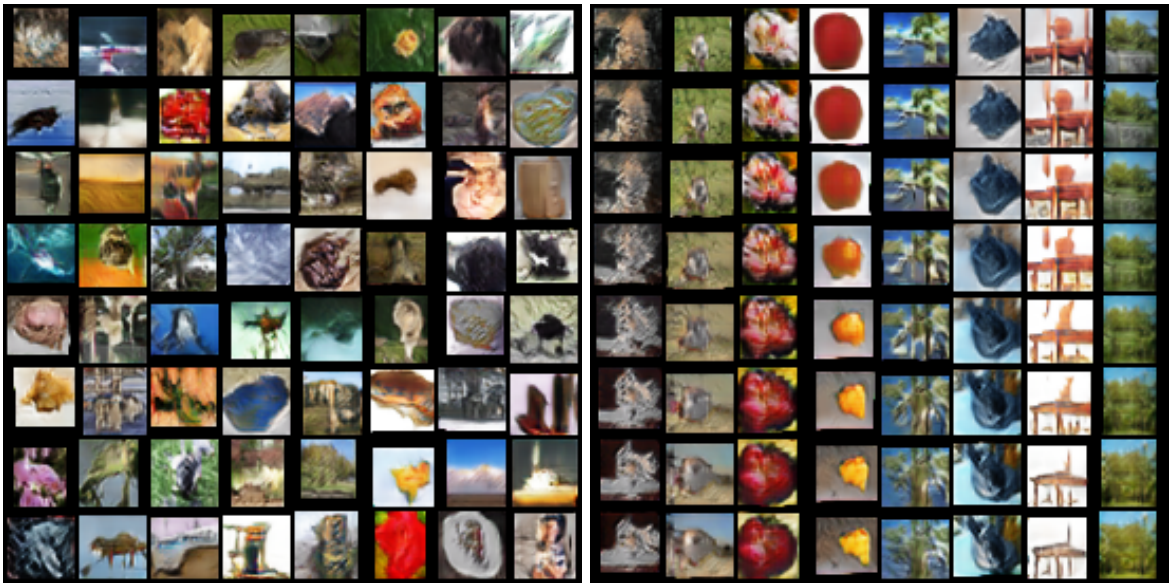
## Results

The trained model produces class-conditional images that exhibit coherent global structure and recognisable object forms across a wide range of categories. A randomly sampled batch of generated images shows clear variation in colour, shape, and background statistics, indicating that the model does not collapse to a small set of modes.

Quantitatively, the model achieves a Fréchet Inception Distance of 28.6 when comparing 10,000 generated samples to the CIFAR-100 test set. This score is consistent with the observed visual quality and reflects a balance between sample realism and diversity at this model scale.

Latent-space interpolations between pairs of noise vectors produce smooth and visually plausible transitions. Semantic attributes such as colour and object structure evolve gradually along the interpolation path, suggesting that the generator has learned a reasonably continuous latent representation rather than relying on discrete memorisation.

Overall, the qualitative and quantitative results indicate that the model captures meaningful class-conditional structure while maintaining diversity across generated samples.



## Limitations

Despite producing coherent images, the model exhibits limitations in fine-grained detail. Textures and small structural elements are often less sharp than in real images, particularly for visually complex classes. This reflects the trade-off between global coherence and local detail inherent in a compact generator architecture.

Diversity across classes is present but uneven. Some categories display reduced variation in pose or background, suggesting that the generator allocates capacity preferentially to dominant visual features rather than fully modelling all modes of the data distribution. This effect is more pronounced for classes with high intra-class variability.

While latent-space interpolations are generally smooth, certain transitions reveal ambiguity in semantic structure, especially when interpolating between visually dissimilar samples. These behaviours highlight constraints in the learned representation and reflect the balance between stability, expressiveness, and model scale.

Taken together, these limitations arise from the interaction between model capacity, training duration, and dataset complexity, and illustrate the trade-offs involved in conditional generative modelling at this scale.

# References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